

PA 542: Advanced Data Analysis II

Spring 2025

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Time and Location: Tuesday 2:00pm – 4:50pm, 2201 AEH

Instructor: Michael D. Siciliano (sicilian@uic.edu), 2116 AEH

Instructor Office Hours: TBA (but I'm happy to meet any time; just email me)

Teaching Assistant: Shuwen Zhang

Teaching Assistant Office Hours: Tuesdays 12 pm to 1 pm via Zoom. Email to set up other times.

I. LEARNING AND COURSE OBJECTIVES:

This course builds on a foundation in statistics, linear regression, and data analysis. The course covers a series of topics and methods critical to the analysis of social science data, including count regression models, regression discontinuity designs, instrumental variables, matching methods, multilevel models, and longitudinal multilevel models. This course emphasizes conceptual understanding and the appropriate application of methods given one's sampling strategy, data, and theory. The course will entail a series of problem sets to help students become familiar with the methods/designs and get their hands dirty working with data. My goal is that students will not only become proficient in the topics covered in the course but also gain the ability and confidence necessary to go out and learn new methods (i) as needed for their research and (ii) as the field of quantitative analysis progresses.

The course is broken down into three major sections:

- The linear model and its extensions
- Causal inference methods
- Multilevel models

One component of the class will be a replication study where the student recreates the analysis and results of a recently published journal article. I hope you will take your replication paper and extend those results by applying new methodology or adding new variables - this is a great way to produce a quality conference paper and/or journal article.

Students completing this course will:

- Understand the core principles of causal inference, including counterfactuals and the difference between correlation and causation.
- Apply econometric methods such as OLS, IV, DiD, and RDD to estimate causal effects.
- Use R to implement causal inference techniques and analyze real-world data.
- Evaluate the internal and external validity of causal claims in empirical research.
- Formulate and test causal hypotheses using appropriate research designs.
- Clearly communicate causal analysis results through effective visualizations and written reports.
- Understand the conceptual framework of multilevel models, including the distinction between fixed and random effects.
- Specify, estimate, and interpret multilevel models for nested and hierarchical data structures.

II. SOFTWARE REQUIREMENTS

I do not require you to adopt a particular statistical program – if you are proficient in a given software you are free to continue using it for certain aspects of the course. I have worked with SAS and Stata before and have taught courses in different software. However, I conduct nearly all of my work in R and the predecessor of this course, PA 541, was taught in R. While I allow you to work with the software of your choosing, the examples in class and modeling approaches I discuss will be implemented in R. I will also provide tips and tools for better workflow procedures, data handling, and graphics in R. Thus, I would encourage all students, even those who consider themselves proficient in another statistical package to learn and use R. We can discuss software and the limitations of certain packages on the first day of class.

III. READINGS

Each week I will assign a set of readings. This may include methods papers as well as applications. We will use one primary textbook for the causal inference portion of the course. That text is freely available online. For the multilevel modeling section, I will provide copies from selected chapters from two different books. So, **no text is required for purchase**. The primary texts we will draw from are:

- Cunningham, Scott. *Causal Inference: The Mixtape*. Causal Inference. Yale University Press, 2021.
- Hox, Joop J. and Rens van de Schoot. *Multilevel Analysis: Techniques and Applications, Second Edition*.
- Singer, Judith D., and John B. Willett. *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*, 2003.

IV. REQUIREMENTS AND GRADES

1. Replication Paper

During the course of the semester, you will be working on a replication paper. You will work in teams of two. This paper will require you to find a recently published (within the last 5-10 years) article in your field of study and recreate the findings presented in the journal. You will turn this replication in as your final paper. Note: increasingly more journals, especially in economics and political science, require authors to provide replication materials (i.e., the data and the analysis script needed to reproduce the results). You must choose an article that does not provide the script, only the data. Replication papers are due during finals week. Papers should be between 5,000 and 7,500 words, including tables, citations, etc... I will discuss this assignment in more detail in class. The paper you replicate should use one of the methods we discuss in this course or a similar/more advanced method.

2. Problem Sets

Most weeks you will have a problem set to work on. The problem set will focus on applying the methods to real data and interpreting and using the output. I encourage you to work with your classmates on these assignments but to ultimately complete and write up the results individually.

3. New methods presentations

As noted, a major goal of this course is for you to gain confidence in your ability to learn and implement new methods. Not only because of the range of methods that exist outside of this course but because quantitative methodology is constantly expanding and evolving (see the difference-in-differences literature over the past few years as an example). Each student, working in teams of two, will be required to present on an advanced methodological topic not covered in this course. You will choose your method in consultation with me and will be required to present the method to the class. You will also prepare a high-level summary of the method for the class (this is a 3-5 page document that discusses the method and how to implement it in R; this could include an example analysis and interpretation of the results). Some options include ordinal/multinomial logistic regression,

spatial analysis, path models/SEM, qualitative comparative analysis, survival analysis, regression trees/splines, meta-analysis, machine learning topics, etc. Presentations will be given during the last weeks of class.

4. Midterm

There will be an in-class midterm. The midterm will cover material from all preceding weeks. The midterm will focus on the student's ability to interpret model results. It will be a closed-book exam.

5. MLM Quiz

There will be a short quiz on multilevel models given at the end of the semester.

Your grade will be calculated as follows:

Component	Percentage of overall grade	Due date
Replication Paper	25%	May 6
Problem Sets	25%	To be scheduled
New Methods Presentation	10%	April 22 & 29
Midterm	35%	March 18
MLM Quiz	5%	April 22

V. WEEKLY SCHEDULE (readings subject to change)

PART ONE – THE LINEAR MODEL AND ITS EXTENSIONS

Session 1 – January 14

Topics: Course overview and syllabus. Regression Review. Linear models, qualitative predictors, interaction effects.

Reading:

King, Gary (2006) Publication, *PS: Political Science and Politics*, Vol. XXXIX, No. 1, 119-125 [This provides some insight on how you can extend your replication paper into a published work]

Session 2 – January 21

Topics: Generalized linear models and logistic regression. Data wrangling.

Readings:

Çetinkaya-Rundel and Johanna Hardin (2021) Introduction to Modern Statistics, First Edition. Chapter on Logistic Regression: <https://openintro-ims.netlify.app/model-logistic.html>

dplyr tutorial (optional; just for those needing a refresher):
<https://dplyr.tidyverse.org/articles/dplyr.html>

Session 3 – January 28 [Async Session]

Topics: Count Regression models (Poisson, negative binomial, zero inflated). Maximum likelihood estimation.

Readings:

Atkins, David C. and Gallop, Robert J. (2007) Rethinking how family researchers model infrequent outcomes: A tutorial on count regression and zero-inflated models.

Coxe, S., West, S. G., & Aiken, L. S. (2009). The Analysis of Count Data: A Gentle Introduction to Poisson Regression and its Alternatives. *Journal of Personality Assessment*, 91(2), 121-136.

PART TWO – CAUSAL INFERENCE

Session 4 – February 4

Topics: Prediction versus causation. Directed acyclic graphs (DAGs). Sensitivity analysis.

Readings:

Cunningham Book Chapters: Introduction, Directed Acyclic Graphs, Potential Outcomes Causal Model

Optional Readings:

Rohrer, J. M. (2018). Thinking Clearly About Correlations and Causation: Graphical Causal Models for Observational Data. *Advances in Methods and Practices in Psychological Science*, 1(1), 27-42. doi:10.1177/2515245917745629

Shrier, I., & Platt, R. W. (2008). Reducing bias through directed acyclic graphs. *BMC Medical Research Methodology*, 8(1), 70. doi:10.1186/1471-2288-8-70

Useful resources and a tutorial for the online program *dagitty*: <http://dagitty.net/learn/>

Session 5 – February 11

Topics: Matching

Reading:

Cunningham Book Chapter: Matching and Subclassification

Optional Reading:

Stuart, E. A. (2010). Matching Methods for Causal Inference: A Review and a Look Forward. *Statistical Science*, 25(1), 1–21, 21.

Session 6 – February 18

Topics: Regression Discontinuity

Readings:

Cunningham Book Chapter: Regression Discontinuity

Optional Reading:

Lee, D. S., & Lemieux, T. (2010). Regression Discontinuity Designs in Economics. *Journal of Economic Literature*, 48(2), 281–355. <https://doi.org/10.1257/jel.48.2.281>

Session 7 – February 25

Topics: Instrumental Variables. Using instruments in RD.

Readings:

Cunningham Book Chapter: Instrumental Variables

Optional Readings:

Angrist, J. D., & Krueger, A. B. (2001). Instrumental Variables and the Search for Identification: From Supply and Demand to Natural Experiments. *The Journal of Economic Perspectives*, 15(4), 69-85.

Bennett, D. A. (2010). An Introduction to Instrumental Variables Analysis: Part 1. *Neuroepidemiology*, 35(3), 237-240.

Session 8 – March 4

Topics: Difference-in-Differences.

Readings:

Cunningham Book Chapters: Panel Data (just skim), Difference-in-Differences

Optional Readings:

Dougherty, Christopher. (2011) "Introduction to Econometrics", 4th ed. Oxford University Press: New York.
Chapter 14 - Introduction to Panel Methods.

Angrist, Joshua D., and Jörn-Steffen Pischke. Mastering' metrics: The path from cause to effect. Princeton University Press, 2014. Chapter 5: Difference-in-Differences.

Session 9 – March 11 [Online Session]

Topics: Brief intro on synthetic control. Review for midterm.

Readings:

Cunningham Book Chapter: Synthetic Control

Midterm Exam – March 18

Topics: **In Class Midterm

NO CLASS – March 25

UIC Spring Break

PART THREE – MULTILEVEL MODELING

Session 10 – April 1

Topics: Multilevel Models (Hierarchical Linear Models) part 1

Readings:

Hox, Joop et al. "Multilevel Analysis: Techniques and Applications", Chapter 2.

Hofmann, D. A. (1997). An Overview of the Logic and Rationale of Hierarchical Linear Models. *Journal of Management*, 23(6), 723-744.

Session 11 – April 8

Topics: Multilevel Models (Hierarchical Linear Models) part 2

Reading:

Cohen et al. (2003) "Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences. 3rd edition. Chapter 14: Random Coefficient Regression and Multilevel Models.

Optional Readings:

Hayes, A. F. (2006). A Primer on Multilevel Modeling. *Human Communication Research*, 32(4), 385-410.

Enders, C. K., & Tofighi, D. (2007). Centering predictor variables in cross-sectional multilevel models: A new look at an old issue. *Psychological Methods*, 12(2), 121-138.

Session 12 – April 15 [Online Session]

Topics: Introduction to Longitudinal Multilevel Models. Class time to work on replication papers and new method presentations.

Methods Readings:

Singer, Judith D. and Willett, John B. (2003). "Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence." Oxford University Press: New York. Chapters 1-4

Quiz and Presentations – April 22nd

Topics: Short quiz on multilevel models.

****Due:** New Methods Presentations

Presentations – April 29

****Due:** New Methods Presentations

Finals Week – May 6

****Due:** Replication Paper

VI. COURSE AND UNIVERSITY POLICIES

Academic Integrity: As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students shall be handled pursuant to the Student Disciplinary Policy. The Student Disciplinary Policy is available online at <https://dos.uic.edu/wp-content/uploads/sites/262/2018/10/DOS-Student-Disciplinary-Policy-2018-2019-FINAL.pdf>.

Special Needs: UIC and the Department of Public Policy, Management, and Analytics, are committed to maintaining a barrier-free environment so individuals with disabilities can fully access programs, services and all activities on campus. The Office of Disability Services works to ensure the accessibility of UIC programs, classes, and services to students with disabilities. Services are available for students who have documented disabilities, including vision or hearing impairments and emotional or physical disabilities. Students with disability/access needs or questions may contact the Office of Disability Services at (312) 413-2183 (voice) or (312) 413-0123 (TTY only). Please feel free to contact me if you need any special accommodations.

Diversity and Inclusion: It is my goal that people from diverse backgrounds and perspectives be included in and served by this course, the Department of Public Policy, Management, and Analytics, and the University of Illinois Chicago. I believe that the diversity that the students bring to this class is a resource that can be used to improve student learning and perspectives. Course materials and activities are designed to respect all types of diversity: gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, values, and culture. I intend this class to be an environment where all students feel safe to express their perspectives and opinions. Please let me know if something said or done by me, the TA, guest lecturers, or other students, is troubling or causes discomfort or offense. If this occurs, please feel free to discuss the situation with me privately, to raise your concern in class, or to let me know about the issue through a trusted source (e.g., another student, faculty member, or your academic advisor).

Campus Advocacy Network: Under the Title IX law you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim-services

and advocacy contact UIC's Campus Advocacy Network at 312-413-1025 or visit <http://can.uic.edu/>. To make a report to UIC's Title IX office, contact Rebecca Gordon, EdD at TitleIX@uic.edu or (312) 996-5657.